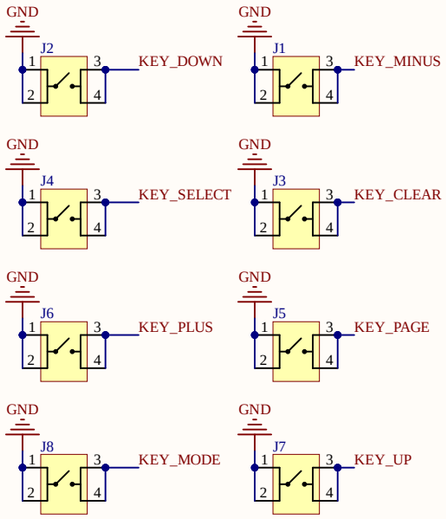
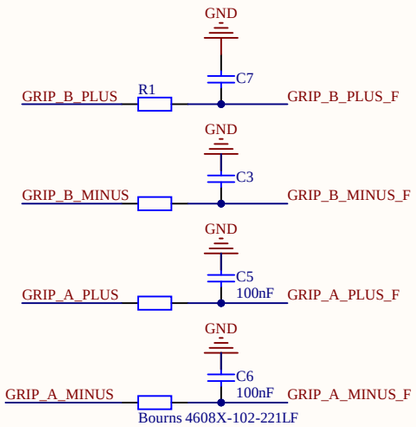
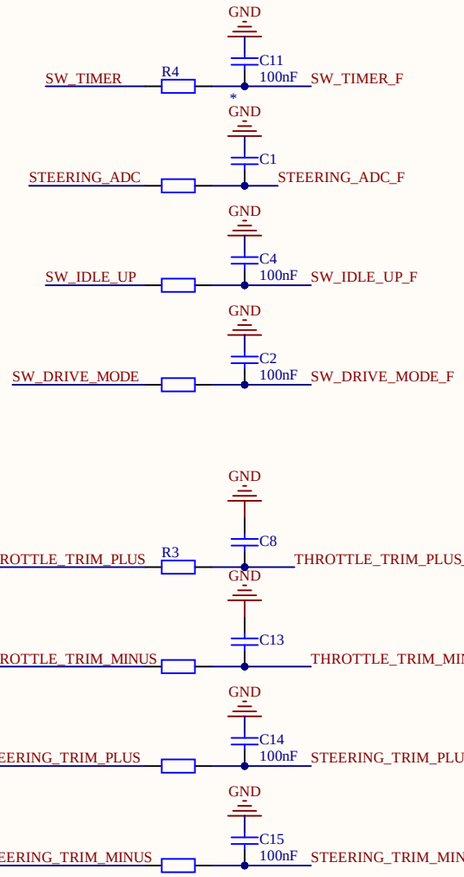
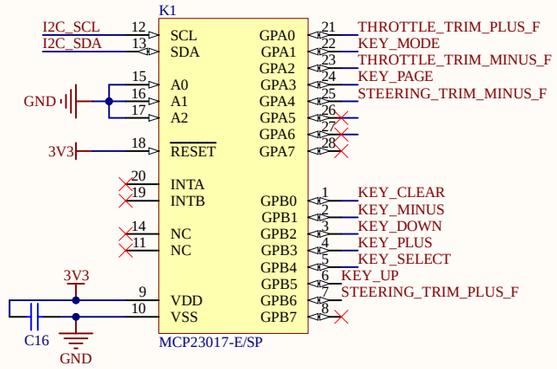
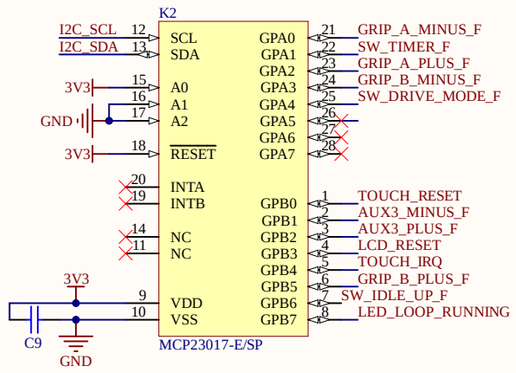
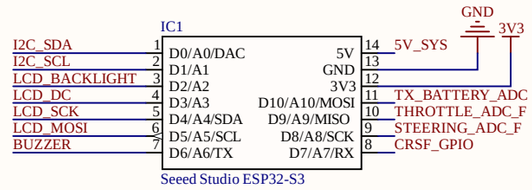

GND
J2
KEY_DOWN
J1
KEY_MINUS
J4
KEY_SELECT
J3
KEY_CLEAR
J6
KEY_PLUS
J5
KEY_PAGE
J8
KEY_MODE
J7
KEY_UP

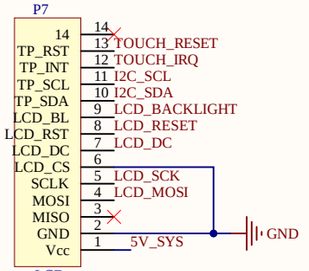

GRIP_B_PLUS
R1
C7
GRIP_B_PLUS_F
GRIP_B_MINUS
C3
GRIP_B_MINUS_F
GRIP_A_PLUS
C5
100nF
GRIP_A_PLUS_F
GRIP_A_MINUS
C6
100nF
GRIP_A_MINUS_F
Bourns 4608X-102-221LF

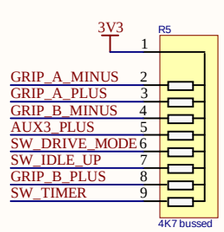

SW_TIMER
R4
C11
100nF
SW_TIMER_F
STEERING_ADC
C1
STEERING_ADC_F
SW_IDLE_UP
C4
100nF
SW_IDLE_UP_F
SW_DRIVE_MODE
C2
100nF
SW_DRIVE_MODE_F
THROTTLE_TRIM_PLUS
R3
C8
THROTTLE_TRIM_PLUS_F
THROTTLE_TRIM_MINUS
C13
THROTTLE_TRIM_MINUS_F
STEERING_TRIM_PLUS
C14
100nF
STEERING_TRIM_PLUS_F
STEERING_TRIM_MINUS
C15
100nF
STEERING_TRIM_MINUS_F

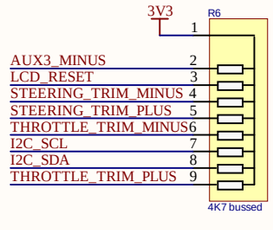

K1
I2C_SCL
I2C_SDA
SCL
SDA
A0
A1
A2
RESET
INTA
INTB
NC
NC
VDD
VSS
GPA0
GPA1
GPA2
GPA3
GPA4
GPA5
GPA6
GPA7
GPB0
GPB1
GPB2
GPB3
GPB4
GPB5
GPB6
GPB7
THROTTLE_TRIM_PLUS_F
KEY_MODE
THROTTLE_TRIM_MINUS_F
KEY_PAGE
STEERING_TRIM_MINUS_F
KEY_CLEAR
KEY_MINUS
KEY_DOWN
KEY_PLUS
KEY_SELECT
KEY_UP
STEERING_TRIM_PLUS_F
3V3
GND
C16
MCP23017-E/SP

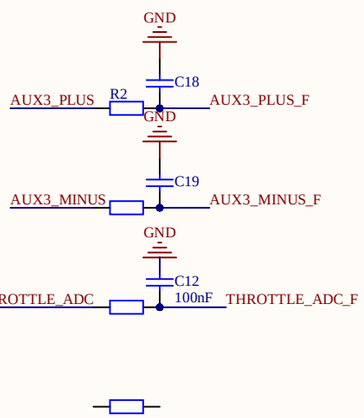

K2
I2C_SCL
I2C_SDA
SCL
SDA
A0
A1
A2
RESET
INTA
INTB
NC
NC
VDD
VSS
GPA0
GPA1
GPA2
GPA3
GPA4
GPA5
GPA6
GPA7
GPB0
GPB1
GPB2
GPB3
GPB4
GPB5
GPB6
GPB7
GRIP_A_MINUS_F
SW_TIMER_F
GRIP_A_PLUS_F
GRIP_B_MINUS_F
SW_DRIVE_MODE_F
TOUCH_RESET
AUX3_MINUS_F
AUX3_PLUS_F
LCD_RESET
TOUCH_IRQ
GRIP_B_PLUS_F
SW_IDLE_UP_F
LED_LOOP_RUNNING
3V3
GND
C9
MCP23017-E/SP

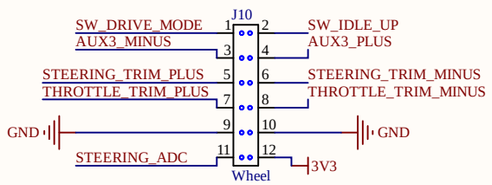

IC1
I2C_SDA
I2C_SCL
LCD_BACKLIGHT
LCD_DC
LCD_SCK
LCD_MOSI
BUZZER
D0/A0/DAC
D1/A1
D2/A2
D3/A3
D4/A4/SDA
D5/A5/SCL
D6/A6/TX
5V
GND
3V3
D10/A10/MOSI
D9/A9/MISO
D8/A8/SCK
D7/A7/RX
5V_SYS
TX_BATTERY_ADC
THROTTLE_ADC_F
STEERING_ADC_F
CRSF_GPIO
Seeed Studio ESP32-S3

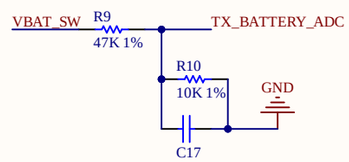

P7
TP_RST
TP_INT
TP_SCL
TP_SDA
LCD_BL
LCD_RST
LCD_DC
LCD_CS
SCLK
MOSI
MISO
GND
Vcc
TOUCH_RESET
TOUCH_IRQ
I2C_SCL
I2C_SDA
LCD_BACKLIGHT
LCD_RESET
LCD_DC
LCD_SCK
LCD_MOSI
5V_SYS
GND
LCD

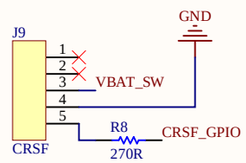

3V3
R5
GRIP_A_MINUS
GRIP_A_PLUS
GRIP_B_MINUS
AUX3_PLUS
SW_DRIVE_MODE
SW_IDLE_UP
GRIP_B_PLUS
SW_TIMER
4K7 bussed


3V3
R6
AUX3_MINUS
LCD_RESET
STEERING_TRIM_MINUS
STEERING_TRIM_PLUS
THROTTLE_TRIM_MINUS
I2C_SCL
I2C_SDA
THROTTLE_TRIM_PLUS
4K7 bussed

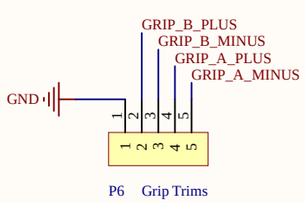

GND
AUX3_PLUS
R2
C18
AUX3_PLUS_F
AUX3_MINUS
C19
AUX3_MINUS_F
THROTTLE_ADC
C12
100nF
THROTTLE_ADC_F

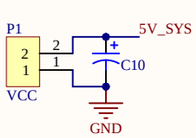

J10
SW_DRIVE_MODE
AUX3_MINUS
STEERING_TRIM_PLUS
THROTTLE_TRIM_PLUS
STEERING_ADC
SW_IDLE_UP
AUX3_PLUS
STEERING_TRIM_MINUS
THROTTLE_TRIM_MINUS
GND
3V3
Wheel

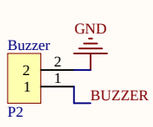

VBAT_SW
R9
47K 1%
TX_BATTERY_ADC
R10
10K 1%
GND
C17
3V3
R11
1K
D1
3V3
R12
1K
D2
RUN
LED_LOOP_RUNNING

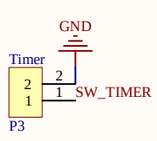

J9
VBAT_SW
R8
270R
CRSF_GPIO
GND
CRSF

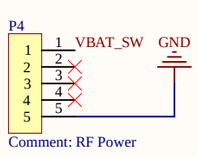

3V3
R7
10K
TOUCH_RESET

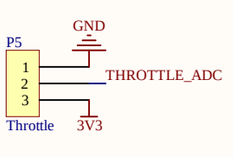

GRIP_B_PLUS
GRIP_B_MINUS
GRIP_A_PLUS
GRIP_A_MINUS
GND
P6 Grip Trims


P1
5V_SYS
C10
VCC
GND


Buzzer
GND
BUZZER
P2


Timer
GND
SW_TIMER
P3


P4
VBAT_SW
GND
Comment: RF Power


P5
GND
THROTTLE_ADC
Throttle
3V3